

## Topic 9 – Ecosystems and material cycles

| Students should: |                                                                                                                                                                                                                                                           | Maths skills                                |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 9.1              | Describe the different levels of organisation from individual organisms, populations, communities, to the whole ecosystem                                                                                                                                 |                                             |
| 9.2              | Explain how communities can be affected by abiotic and biotic factors, including:<br>a temperature, light, water, pollutants<br>b competition, predation                                                                                                  | 4a, 4c                                      |
| 9.3              | Describe the importance of interdependence in a community                                                                                                                                                                                                 |                                             |
| 9.4              | Describe how the survival of some organisms is dependent on other species, including parasitism and mutualism                                                                                                                                             |                                             |
| 9.5              | <i>Core Practical: Investigate the relationship between organisms and their environment using field-work techniques, including quadrats and belt transects</i>                                                                                            | 1c, 1d,<br>2b, 2c, 2d, 2f,<br>2g,<br>4a, 4c |
| 9.6              | Explain how to determine the number of organisms in a given area using raw data from field-work techniques, including quadrats and belt transects                                                                                                         | 1c, 1d<br>2b, 2c, 2d, 2g<br>4a, 4c          |
| 9.7B             | Explain how some energy is transferred to less useful forms at each trophic level and that this affects the number of organisms at each trophic level, limits the length of a food chain and determines the shape of a pyramid of biomass in an ecosystem |                                             |
| 9.8B             | Calculate the efficiency of energy transfers between trophic levels and percentage calculations of biomass                                                                                                                                                | 1a, 1b, 1c<br>2c<br>4a                      |
| 9.9              | Explain the positive and negative human interactions within ecosystems and their impacts on biodiversity, including:<br>a fish farming<br>b introduction of non-indigenous species<br>c eutrophication                                                    | 2c, 2g<br>4a, 4c                            |
| 9.10             | Explain the benefits of maintaining local and global biodiversity, including the conservation of animal species and the impact of reforestation                                                                                                           |                                             |

| Students should:                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Maths skills                    |
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| <p>9.11B Describe the biological factors affecting levels of food security, including:</p> <ul style="list-style-type: none"> <li>a increasing human population</li> <li>b increasing animal farming and the increased meat and fish consumption</li> <li>c the impact of new pests and pathogens</li> <li>d environmental change caused by human activity</li> <li>e sustainability issues, e.g. use of land for biofuel production and the cost of agricultural inputs</li> </ul> | <p>2c<br/>4a, 4c</p>            |
| <p>9.12 Describe how different materials cycle through the abiotic and biotic components of an ecosystem</p>                                                                                                                                                                                                                                                                                                                                                                        |                                 |
| <p>9.13 Explain the importance of the carbon cycle, including the processes involved and the role of microorganisms as decomposers</p>                                                                                                                                                                                                                                                                                                                                              |                                 |
| <p>9.14 Explain the importance of the water cycle, including the processes involved and the production of potable water in areas of drought including desalination</p>                                                                                                                                                                                                                                                                                                              |                                 |
| <p>9.15 Explain how nitrates are made available for plant uptake, including the use of fertilisers, crop rotation and the role of bacteria in the nitrogen cycle</p>                                                                                                                                                                                                                                                                                                                |                                 |
| <p>9.16B <b>Evaluate the use of indicator species as evidence to assess the level of pollution, including:</b></p> <ul style="list-style-type: none"> <li><b>a polluted water – bloodworm, sludgeworm</b></li> <li><b>b clean water – freshwater shrimps, stonefly</b></li> <li><b>c air quality – different species of lichen, blackspot fungus on roses</b></li> </ul>                                                                                                            | <p>2c, 2g<br/>4a, 4c</p>        |
| <p>9.17B Explain the effects of temperature, water content and oxygen availability on the rate of decomposition in food preservation</p>                                                                                                                                                                                                                                                                                                                                            |                                 |
| <p>9.18B Explain the effects of temperature, water content and oxygen availability on the rate of decomposition in composting</p>                                                                                                                                                                                                                                                                                                                                                   | <p>2c<br/>4a, 4c</p>            |
| <p>9.19B Calculate rate changes in the decay of biological material</p>                                                                                                                                                                                                                                                                                                                                                                                                             | <p>1c<br/>2c, 2f<br/>4a, 4c</p> |

**Use of mathematics**

- Calculate surface area : volume ratios (1c).
- Plot, draw and interpret appropriate graphs (4a, 4b, 4c and 4d).
- Understand and use percentiles and calculate percentage gain and loss of mass (1c).
- Translate information between numerical and graphical forms (4a).
- Construct and interpret frequency tables and diagrams, bar charts and histograms (2c).
- Understand the principles of sampling as applied to scientific data (2d).
- Use a scatter diagram to identify a correlation between two variables (2g).
- Calculate the percentage of mass (1c).
- Use fractions and percentages (1c).
- Calculate arithmetic means (2b).
- Calculate the rate changes in the decay of biological material (1c).
- Extract and interpret information from charts, graphs and tables (2c, 4a).

**Suggested practicals**

- Investigate tropic responses.
- Investigate how indicator species can be used to assess levels of pollution in water or the atmosphere.
- Investigate animal behaviour using choice chambers.